Lab 4: Liquid Chemistry

BIOL-1

Name: ____________________	**Date:** ____________________

Goal: Perform simplified experiments to observe diffusion, viscosity, redox reactions, and serial dilutions using minimal reagents.

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1. Diffusion and Mixing (Color in Water vs. Salt Water)

Goal: See how dyes spread in still water vs. salt solution.

Materials

- 2–3 test tubes
- Distilled water
- 5% NaCl solution
- Red food color (or methylene blue as a second dye, optional)

Procedure

1. **Fill tube A:** ¾ full with distilled water.
2. **Fill tube B:** ¾ full with 5% NaCl solution.
3. Use a straw or improvised dropper to place **one** small drop of red food color at the surface of tube A. **Do not stir.**
4. Repeat with tube B (same drop size, same position).
5. Place the tubes side by side in the rack, don't bump them, and watch for 5–15 minutes.

What to Look For

- The colored drop slowly spreads out in each tube until the color is more uniform (**diffusion**).
- In the salt solution, spreading is often slower, and the color boundary can look sharper or "hang" longer.

Tip: You can reuse these tubes later by diluting further with water so you don't burn through dye.

2. Viscosity with Methyl Cellulose (Falling or Rising Drops)

Goal: Compare dye motion in water vs. methyl cellulose (thick solution).

Materials

- 2 test tubes
- Distilled water
- Methyl cellulose solution
- Methylene blue (or red food color)

Procedure

1. **Fill tube A:** ½–¾ full of distilled water.
2. **Fill tube B:** Same volume of methyl cellulose solution.
3. Put a small drop of methylene blue at the top of tube A; **avoid stirring.**
4. Put the same size drop at the top of tube B.
5. Watch for several minutes.

Optional variation: In a second run, place the drops near the bottom of each tube instead and watch them rise.

What to Look For

- **In water:** The drop stretches and mixes quickly; the color spreads out rapidly.
- **In methyl cellulose:** The drop moves and mixes more slowly and keeps its shape longer, showing the effect of higher **viscosity**.

Tip: You can do this right after Experiment 1, using the same dyes; just pour small amounts into clean tubes if you are low on glassware.

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3. Methylene Blue + Bleach (Simple Redox / Bleaching)

Goal: Show that an oxidizer (bleach) destroys the color of methylene blue.

Safety

- **Wear eye protection and gloves** when handling 10% bleach.
- Work with drops, not large volumes.

Materials

- 2–3 test tubes
- Methylene blue solution (make it just visibly blue, not dark)
- 10% bleach
- Distilled water

Procedure (Single-Tube Version)

1. Fill a test tube ½ full with distilled water.
2. Add methylene blue dropwise until it is a clear, medium blue.
3. Add 1 drop of bleach, gently swirl.
4. Continue adding bleach one drop at a time, swirling after each, and note the number of drops until the solution is nearly colorless.

Simple Multi-Tube Variant

1. Prepare a small beaker or cup with light-blue methylene blue solution.
2. Fill 3 test tubes ¼–½ full from this same beaker so all start with the same color.
3. **Tube 1:** Add 1 drop bleach, swirl.
4. **Tube 2:** Add 3 drops bleach.
5. **Tube 3:** Add 5 drops bleach.

Result: You get a little "scale" of decreasing blue to nearly colorless.

What to Look For

- The blue color fades as hypochlorite in bleach oxidizes and destroys the conjugated structure of methylene blue.

Tip: You can reserve one of the less-bleached tubes to compare later with permanganate solutions in Experiment 4.

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4. Potassium Permanganate Serial Dilution (Color and Concentration)

Goal: Make a color gradient to illustrate concentration; uses very little $KMnO_4$.

Safety

- Use just a **tiny** crystal of $KMnO_4$; it stains skin and surfaces.
- **Wear gloves**; rinse spills with plenty of water.

Materials

- 4–5 small test tubes (if you have fewer, use 3; the idea is the same)
- Beaker
- Potassium permanganate crystals
- Distilled water

Procedure

1. Put ~50–100 mL distilled water in a beaker.
2. Add the smallest visible crystal of $KMnO_4$ and stir until the solution is a deep but transparent purple. **This is your stock.**
3. Label tubes 1–4 (or 1–5).
4. Add the same volume of distilled water to each tube (for example, 5 mL).
5. Using a clean dropper or straw:
    - Add 1 mL (or a few drops) of stock to **Tube 1** and mix.
    - Take 1 mL from **Tube 1** and transfer to **Tube 2**, mix.
    - Take 1 mL from **Tube 2** and transfer to **Tube 3**, mix.
    - Continue for **Tube 4** (and 5 if available).

What to Look For

- **Tube 1** will be darkest; each subsequent tube is lighter but still clearly colored.
- Even in the palest tube there is visible purple, illustrating that a very small initial amount contains a huge number of particles.

Variation: If tubes are limited, you can instead make just three tubes with "dark," "medium," and "light" by adding different numbers of drops of the stock to fixed volumes of water.